

Kevin R Sitek

Research Assistant Professor, Northwestern University

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Education

- PhD** **Harvard University**, Speech and Hearing Bioscience and Technology 2013 – 2019
- Division of Medical Sciences, Harvard–MIT Program
 - Advisor: Satrajit S. Ghosh, PhD
 - Dissertation: [Mapping the Human Subcortical Auditory System With MRI](#)
- BA** **University of California, Berkeley**, Linguistics (Honors) and Cognitive Science 2007 – 2011
- Advisor: Keith Johnson, PhD
 - Linguistics honors thesis: Phonetic context effects in ipsilateral and contralateral conditions

Experience

- Northwestern University**, Research Assistant Professor Evanston, IL 2023 – present
- Department of Communication Sciences and Disorders
 - Affiliated Faculty: Cognitive Science Program; Northwestern University Interdepartmental Neuroscience (NUIN) PhD Program
- University of Pittsburgh**, Research Scientist Pittsburgh, PA 2021 – 2023
- Department of Communication Science and Disorders
 - SoundBrain Lab (PI: Bharath Chandrasekaran, PhD)
- University of Pittsburgh**, Postdoctoral Associate Pittsburgh, PA 2021 – 2021
- Department of Communication Science and Disorders
 - SoundBrain Lab (PI: Bharath Chandrasekaran, PhD)
- Baylor College of Medicine**, Postdoctoral Associate Houston, TX 2019 – 2021
- Department of Neuroscience
 - High Resolution Brain Imaging Lab (PI: David Ress, PhD)
- MIT McGovern Institute for Brain Research**, Graduate Student Researcher Cambridge, MA 2014 – 2019
- Gabrieli Laboratory
- San Francisco VA Medical Center / NCIRE**, Staff Research Assistant I/II San Francisco, CA 2011 – 2013
- Brain Imaging and EEG Laboratory (PIs: Judith Ford, PhD; Daniel Mathalon, PhD, MD)
- UC Berkeley Department of Linguistics**, Undergraduate/Staff Research Assistant Berkeley, CA 2009 – 2011
- Phonology Laboratory (PI: Keith Johnson, PhD)

Grants

- Auditory–motor interactions in neural processing and the frequency-following response** 2025–2028
- NIH NIDCD ECR R21 · Grant: 1R21DC022906-01
 - Role: Principal Investigator
 - Co-Investigators: Bharath Chandrasekaran, PhD; Peer Herholz, Dr. rer. nat.

High-resolution functional imaging of speech-induced sensory modulation • NIH NIDCD R01 · Grant: R01 DC020963-01A1 • Role: Co-Investigator (PI: Jason W. Bohland, PhD)	2024–2028
Investigating human non-lemniscal inferior colliculus contributions to auditory learning with 7T MRI • NIH NIDCD K01 Mentored Career Development Award · Grant: K01 DC019421-01A1 • Role: Principal Investigator • Co-Mentors: Bharath Chandrasekaran, PhD; Tamer Ibrahim, PhD; Srivatsun Sadagopan, PhD	2022–2027
Administrative Supplement — K01 DC019421 • NIH NIDCD Administrative Supplement · Grant: 3K01DC019421-05S1 • Role: Principal Investigator	2025–2026
Cortical-subcortical interactions in auditory cognition • CIHR Project Grant • Role: Collaborator (PI: Robert Zatorre, PhD)	2023–2028
Subcortical auditory feedback mechanisms in speech: Function and structure • NIH NIDCD F31 NRSA Predoctoral Fellowship · Grant: F31 DC015695-01 (completed) • Role: Principal Investigator • Sponsor: Satrajit S. Ghosh, PhD; Co-Sponsor: John D. E. Gabrieli, PhD	2016–2018
Submitted Manuscripts and Preprints	
Motor signals modulate cortical but not subcortical processing of self-initiated sounds Raiff, L.†, Butler, G.†, McFarlane, K., Chandrasekaran, B., Sitek, K.R. 10.64898/2026.07.10.737812 (bioRxiv)	in review
Evaluating Approaches for Inference Testing of Whole-Brain Densely Sampled Single-Subject Task fMRI Data Medina, M.C., Reddy, N.H., Bright, M.G., Sitek, K.R. 10.64898/2026.06.29.735344 (bioRxiv)	in review
Distinct 7T functional MRI response patterns to stimulus, response, and feedback across dorsal striatum during auditory learning Sitek, K.R., Roark, C.L., Helou, L.B., Chandrasekaran, B.	in revision
The recognition and comprehension of speech sound patterns show a mechanistic dissociation along the ventral and dorsal streams Llanos, F.†, Sitek, K.R.†, Feng, G., Chandrasekaran, B.	in revision
Auditory corticostriatal connections in the human brain Sitek, K.R., Helou, L.B., Chandrasekaran, B. 10.1101/2022.08.04.502679 (bioRxiv)	in revision
Bayesian Tensor Factorized Mixed Effects Vector Autoregressive Processes for Inferring Granger Causality Patterns from High-Dimensional Neuroimage Data Fan, J., Sitek, K.R., Chandrasekaran, B., Sarkar, A. 10.48550/arXiv.2206.10757 (arXiv)	2022

Publications

- Mapping Whole-Brain Auditory Activation with 3T Multi-Echo fMRI at the Group and Individual-Subject Level** 2026
Medina, M.C., Reddy, N.H., **Sitek, K.R.**, Bright, M.G.
[10.1016/j.heares.2026.109622](https://doi.org/10.1016/j.heares.2026.109622) (Hearing Research)
- Communication is the foundation of an impactful and resilient scientific community** 2025
Sitek, K.R., Wearn, A., DuPre, E.M.
[10.52294/001c.143510](https://doi.org/10.52294/001c.143510) (Aperture Neuro 2025;5(SI 3))
- The Organization for Human Brain Mapping Time Machine: A freely accessible archive of Annual Meeting talks on YouTube** 2025
Wearn, A., **Sitek, K.R.**, Valk, S.L., Forkel, S.J.
[10.52294/001c.138647](https://doi.org/10.52294/001c.138647) (Aperture Neuro 2025;5(SI 2))
- Functional connectivity across the human subcortical auditory system using an autoregressive matrix-Gaussian copula graphical model approach with partial correlations** 2024
Chandra, N.K.†, **Sitek, K.R.**†, Sarkar, A., Chandrasekaran, B.
[10.1162/imag_a_00258](https://doi.org/10.1162/imag_a_00258) (Imaging Neuroscience)
- Cortical-subcortical interactions to violations of auditory predictions measured with 7T functional MRI** 2024
Ara, A., Provias, V., **Sitek, K.R.**, Coffey, E., Zatorre, R.
[10.1093/cercor/bhae316](https://doi.org/10.1093/cercor/bhae316) (Cerebral Cortex)
- On the role of neural oscillations across timescales in speech and music processing** 2022
Gnanateja, G.N., Devaraju, D.S., Heyne, M., Quique, Y.M., **Sitek, K.R.**, Tardif, M.C., Tessmer, R., Dial, H.R.
[10.3389/fncom.2022.872093](https://doi.org/10.3389/fncom.2022.872093) (Frontiers in Computational Neuroscience)
- Structural connectivity of human inferior colliculus subdivisions using in vivo and post mortem diffusion MRI tractography** 2022
Sitek, K.R., Calabrese, E., Johnson, G.A., Ghosh, S.S., Chandrasekaran, B.
[10.3389/fnins.2022.751595](https://doi.org/10.3389/fnins.2022.751595) (Frontiers in Neuroscience)
- Depth relationships and measures of tissue thickness in dorsal midbrain** 2020
Truong, P., Kim, J.H., Savjani, R., **Sitek, K.R.**, Hagberg, G., Scheffler, K., Ress, D.
[10.1002/hbm.25185](https://doi.org/10.1002/hbm.25185) (Human Brain Mapping)
- Mapping the human subcortical auditory system using histology, postmortem MRI and in vivo MRI at 7T** 2019
Sitek, K.R.†, Gulban, O.F.†, Calabrese, E., Johnson, G.A., Lage-Castellanos, A., Moerel, M., Ghosh, S.S.^, De Martino, F.^
[10.7554/eLife.48932](https://doi.org/10.7554/eLife.48932) (eLife)
- Alterations in the inferior longitudinal fasciculus in autism and associations with visual processing: a diffusion-weighted MRI study** 2018
Boets, B., Van Eylen, L., **Sitek, K.R.**, Moors, P., Noens, I., Steyaert, J., Sunaert, S., Wagemans, J.
[10.1186/s13229-018-0188-6](https://doi.org/10.1186/s13229-018-0188-6) (Molecular Autism)

Diffusion tensor imaging of central auditory pathways in patients with sensorineural hearing loss: a systematic review	2018
Tarabichi, O., Kozin, E.D., Kanumuri, V.V., Barber, S., Ghosh, S., Sitek, K.R. , Reinshagen, K., Herrmann, B., Remenschneider, A.K., Lee, D.J. 10.1177/0194599817739838 (Otolaryngology–Head and Neck Surgery)	
Decreased cerebellar-orbitofrontal connectivity correlates with stuttering severity: whole-brain functional and structural connectivity associations with persistent developmental stuttering	2016
Sitek, K.R. , Cai, S., Beal, D.S., Perkell, J.S., Guenther, F.H., Ghosh, S.S. 10.3389/fnhum.2016.00190 (Frontiers in Human Neuroscience)	
Auditory cortex processes variation in our own speech	2013
Sitek, K.R. , Mathalon, D.H., Roach, B.J., Houde, J.F., Niziolek, C.A., Ford, J.M. 10.1371/journal.pone.0082925 (PLoS ONE)	

Teaching

Communicating Science (CSD 550-3)	2024, 2025
• Instructor of Record (Adjunct), Northwestern University • Graduate PhD course: oral, written, visual, and multimedia science communication	
Speech Communication (MIT 6.541/HST.710/SHBT 204)	2015, 2016
• Graduate Teaching Assistant, MIT/Harvard University	
Guest Lectures	2016—present
• Auditory Cognitive Neuroscience — Northwestern CSD 366 (Cognitive Science Proseminar) • Subcortical Auditory Anatomy — Pitt CSD 2110 (Neuroscience of Communication) • Subcortical Contributions to Language — UT Dallas (Brain & Language) • Subcortical Auditory Anatomy and Human Function — Pitt CSD 1024 (Anatomy & Physiology of Hearing) • Acoustic Phonetics: Vowels — Harvard SHBT 204 (Speech Communication)	

Mentorship

PhD Students	
• Laura Raiff, Biomedical Engineering, Northwestern (2026–present; lead advisor, dissertation advisory committee) • Michelle Medina, Biomedical Engineering, Northwestern (2024–present; dissertation advisory committee) • Serena Mon, Communication Sciences and Disorders, Northwestern (2023–2024; qualifying exam committee)	
Doctor of Audiology	
• Laura Cahalan, University of Pittsburgh (Capstone Research Project)	

Undergraduate Research Assistants

- Max Chen, BA Cognitive Science, Northwestern (2025–present; FFRPREP contributor)
- Gabrielle Butler, BA Cognitive Science, Northwestern (2024–2025; honors thesis)
- Amp Kangusmrith, BA Cognitive Science, Northwestern (2024–present; Independent Study in Cognitive Science)
- Angelina Dinardo, BA CSD, University of Pittsburgh (Completed MS in Speech Language Pathology at Northeastern University)
- Madison Andreano, BA CSD, University of Pittsburgh (Completed her MS in Speech Language Pathology at the University of Minnesota)
- Olivia Flemm, BA CSD, University of Pittsburgh (Completed her AuD at the University of Pittsburgh)

High School Research Assistants

- Leo Chen (2025, Northwestern University; FFRPREP contributor)
- Karen Linares Mendoza (2022, University of Pittsburgh; Now pursuing undergraduate studies at Johns Hopkins University)

Awards and Honors

Aperture Neuro Best Reviewer Award	2025
• Organization for Human Brain Mapping, Brisbane, Australia	
ASHA Lessons for Success	2023
• Accepted participant	
OHBM Merit Award	2020
Harvard Brain Science Initiative Young Scientist Travel Award	2018
Mass. Eye and Ear Amelia Peabody Scholarship	2017–2018
OHBM Hackathon Travel Award	2017
• Vancouver, BC	
Neurohackweek / Neurohackademy	2017
• Accepted participant, Seattle, WA	
Certificate of Distinction in Teaching	2015
• Harvard University, Derek Bok Center for Teaching and Learning	

Service

Aperture Neuro Editorial Review Board	2026—present
Frontiers in Language Sciences Review Editor	2022—present
Ad hoc reviewer	
• Journals: <i>Aperture Neuro</i> , <i>Brain & Language</i> , <i>Brain Structure and Function</i> , <i>Cerebral Cortex</i> , <i>Frontiers in Psychology</i> , <i>Heliyon</i> , <i>Journal of the Association for Research in Otolaryngology</i> , <i>Journal of Neuroscience Research</i> , <i>Journal of Speech, Language, and Hearing Research</i> , <i>Molecular Genetics & Genomic Medicine</i> , <i>NeuroImage</i> , <i>Neuropsychologia</i> , <i>OHBM Abstract Submissions</i> , <i>PLOS ONE</i>	
• Granting agencies: European Research Council, Swiss National Science Foundation	
PhD Admissions Committee, Northwestern University Interdepartmental Neuroscience (NUIN) Program	2025–2026

ASHA Convention Topic Committee: Hearing, Balance, Tinnitus	2024, 2025
Undergraduate Studies Committee, Northwestern CSD	2023–2025
Organization for Human Brain Mapping, Communications Committee • Past Chair (2024–2025) • Chair (2023–2024) • Chair-Elect / Blog Team Lead (2022–2023) • Podcast Team Co-founder and Editor (2021–2022)	2021–2025
Organization for Human Brain Mapping, Student–Postdoc Special Interest Group • Social Media and Communications Lead	2019–2021
Society for Cognitive Auditory Neuroscience (SCAN) Moderator	2021–2023
Science in the News (Harvard) • Social Media Outreach Lead (2015–2019) • Editor, Science in the News Blog (2013–2018)	2013–2019
BrainHack Boston Organizing Committee	2017
SMALL Journal Club Co-founder and Organizer	2016–2019